

ANALOG AUDIO SIGNAL PROCESSING CIRCUIT

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INTRODUCTION:

This report focuses on the design, simulation and prototyping of a sound control karaoke circuit. This circuit allows the user to add/subtract the left and right audio channels to create a stereo/karaoke effect. Additionally, it allows the user to adjust the property of sound output such as volume, bass, and treble along with a sound visualizing LED array. The design process starts with modeling the circuit and testing its functionality on NI Multisim, building a breadboard prototype to verify its operation and finally building a PCB prototype.

BLOCK 1: Mixer/Karaoke Module

Design Objective:

The function of this block is to accept the audio input from the 3.5mm audio jack and to either combine it or subtract it for either the mixer mode or karaoke mode. The audio input has 2 channels namely left and right. We need this block to give us an output of $-(L+R)$ for mixer mode and an output of $(L-R)$ for the karaoke mode.

Schematic:

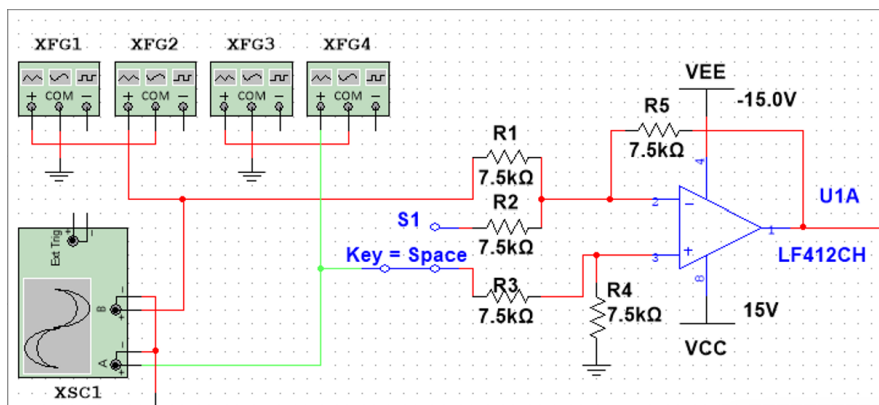


Figure 1, Schematic of Block 1.

Theory of Operation:

The main component of this circuit is the Op AMP, which is being used as both: an inverting summing amplifier and subtracting amplifier. A SPDT switch is used to selectively connect right audio to either the summing node or subtracting node of the op amp. This allows us to choose mixer mode or karaoke mode through the state of the SPDT switch.

Additionally, the karaoke mode works because most audios have their vocals present on both channels and subtracting the right channel from the left cancels out the components of the sound wave that are equally present on both channels.

Derivation/ Analysis:

Mixer Mode:

The Mixer mode deals with an inverting summing amplifier. The standard output for that circuit is

$$v_o = -R_f \left(\frac{v_L}{R_L} + \frac{v_R}{R_R} \right)$$

To achieve our desired gain and output of $-(L + R)$ we need to all of these resistors to be the same value.

$$R_f = R_L = R_R = 7.5 \text{ k}\Omega$$

We selected resistors valued at $7.5 \text{ k}\Omega$ 1% tolerance, to minimize disruptions caused by differing resistor tolerances.

Karaoke Mode:

The karaoke mode deals with a subtracting or difference amplifier. The standard output for that circuit is

$$v_o = \frac{R_2}{R_1} (v_L - v_R)$$

Again, to achieve unity gain and our desired output of $(L - R)$, all the resistor values need to be equal. Since R_2 is already set to $7.5 \text{ k}\Omega$ 1% tolerance from our last step, R_1 also needs to be the same

SIMULATION RESULTS:

For the sake of the simulation the left channel consists of a 0.25 V amplitude 110 Hz sinusoid plus a 0.25 V amplitude 880 Hz sinusoid. The right channel consists of a 0.25 V amplitude 3520 Hz sinusoid plus a 0.25 V amplitude 880 Hz sinusoid. The fact that these input signals have some identical frequency components and some different frequency components allows us to test both the mixer and Karaoke features.

NOTE: The left oscilloscope is the input and the left oscilloscope is the output of block 1. Additionally, in the left oscilloscope, the green signal is the Right Channel while the red signal is the Left Channel.

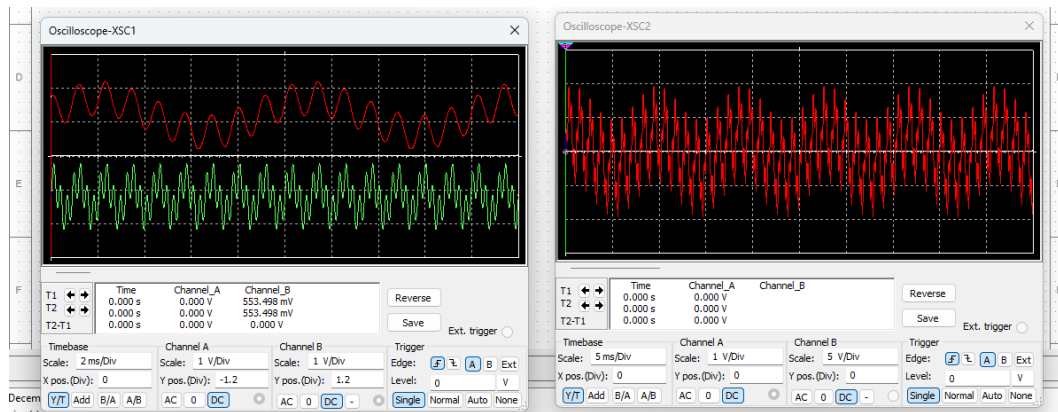


Figure 2, Block 1: Mixer mode

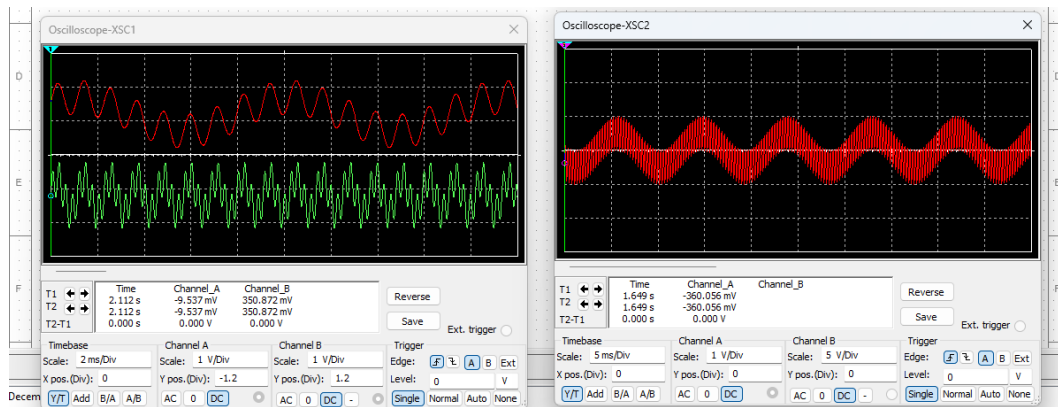


Figure 3, Block 1: Karaoke mode

BLOCK 2: Tone Control Module

Design Objective:

The function of this block is to grant us the tools to be able to modify the bass and treble of the incoming audio signal from block 1. The simplest way to implement this is through the well-known Baxandall tone control circuit. For the purposes of our main circuit, we want to get a gain range of approximately 1/10 to 10 for both bass and treble.

Schematic:

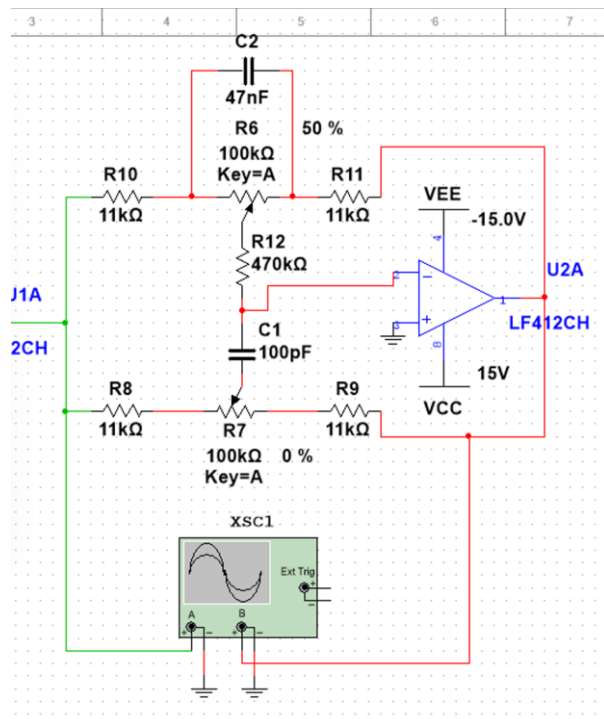


Figure 4, Schematic of Block 2.

Theory of Operation:

The circuit's operation is based on the fact that capacitors act like open circuits at low frequencies and short circuits at high frequencies. Because $C_1 \ll C_2$, it has a higher impedance for any given frequency, so it continues to act like an "open circuit" for higher frequencies than C_2 does. For low input signal frequencies, both capacitors act like open circuits, so the bottom potentiometer setting has no effect on the circuit because the middle terminal is connected to an open circuit. For middle input signal frequencies, the top capacitor (C_2) acts like a short circuit (shorting out the top pot, which is in parallel with the capacitor) and the bottom capacitor (C_1) still acts like an open circuit, so neither potentiometer setting has any effect on the circuit. For high input signal frequencies, both capacitors act like short circuits, so the top potentiometer (shorted out) has no effect on the circuit, but the bottom potentiometer is now connected. The values of the two capacitors and the values of the resistors determine the boundaries between what is considered "low frequency" and what is considered "high frequency." For audio applications, where frequencies range from 20 Hz to 20,000 Hz, the boundary is usually chosen to be in the 400–1000 Hz range.

Derivation/Analysis:

At low frequencies (Bass) the capacitor C_1 behaves like an open circuit. Therefore, the low frequency gain is given by:

$$A_{\text{bass}} = \frac{R_{\text{pot}} + R_{12}}{R_{12}}$$

At high frequencies (Treble) the capacitor C_2 behaves like a short. Therefore, the high frequency gain becomes:

$$A_{\text{treble}} = \frac{R_{\text{pot}} + R_{12}}{R_{12}}$$

At mid-range frequencies one capacitor becomes a short and the other becomes an open circuit. Under these conditions, the gain is balanced by the op amp feedback and becomes unity.

$$A_{\text{mid}} \approx 1$$

Additionally, the max gain of this circuit is

$$A_{V_{\text{max}}} \approx \frac{R_{\text{pot}} + R_s}{R_s}$$

And the min gain is

$$A_{V_{\text{min}}} \approx \frac{R_s}{R_{\text{pot}} + R_s}$$

Where R_s are the series resistors. As we mentioned before, the max and min gain should be 10 and 0.1 respectively. To achieve this, we have chosen the value of R_s to be 11 k Ω .

Simulation Result:

NOTE: The green signal is the input to Block 2 and the red signal is the output of Block 2.

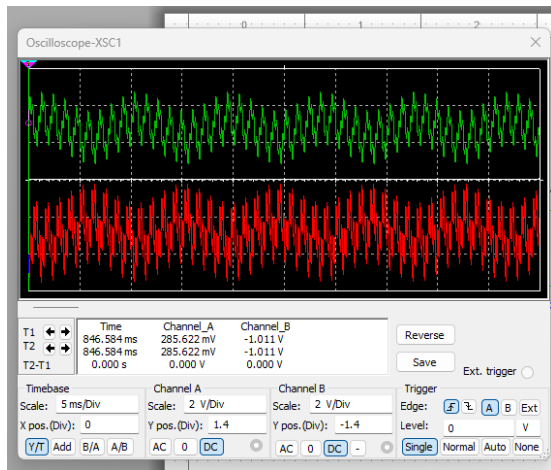


Figure 5, Block 2: Bass boost

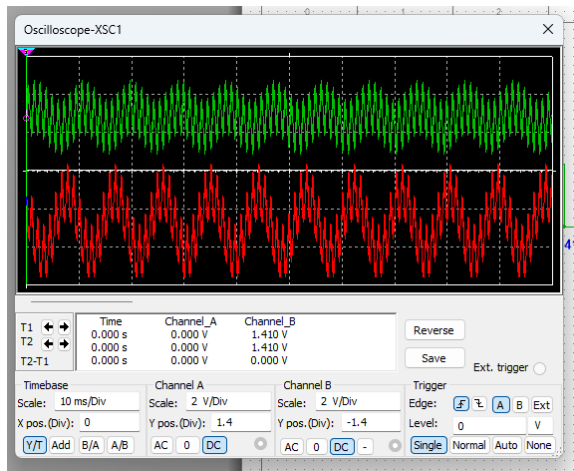


Figure 6, Block 2: Treble boost

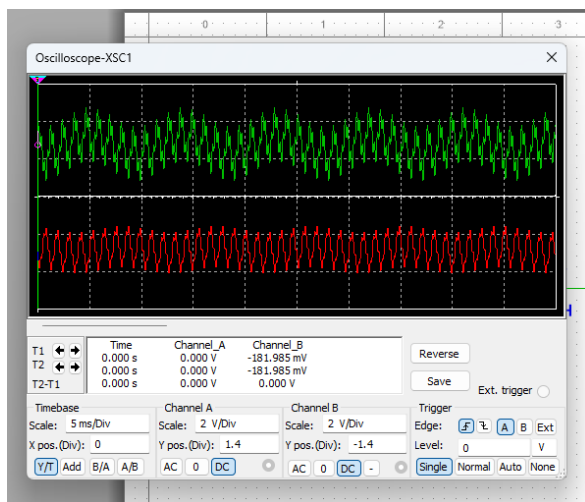


Figure 7, Block 2: Bass and Treble attenuation

BLOCK 3: Volume Control Module

Design Objective:

The objective of this block is very simple: to be able to control the volume of the audio signal from block 2. To achieve this, we will use a single potentiometer.

Schematic:

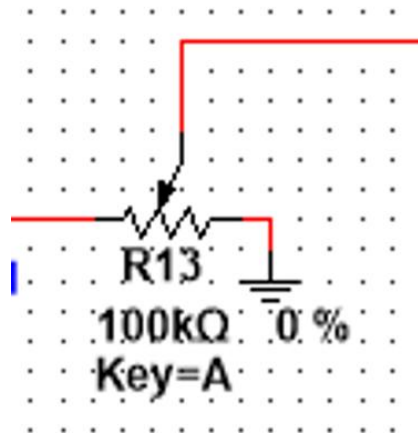


Figure 8: Schematic of Block 3

Theory of Operation:

This block is implemented using a single potentiometer that acts as voltage divider. By adjusting the value of the potentiometer, we are able to output a variable fraction of the input voltage into this block.

Derivation/Analysis:

For a voltage divider the output voltage is given by

$$v_o = v_{in} \frac{R_{wiper}}{R_{pot}}$$

Simulation results:

NOTE: Green channel is the input to the block, while the red channel is the output to the block.

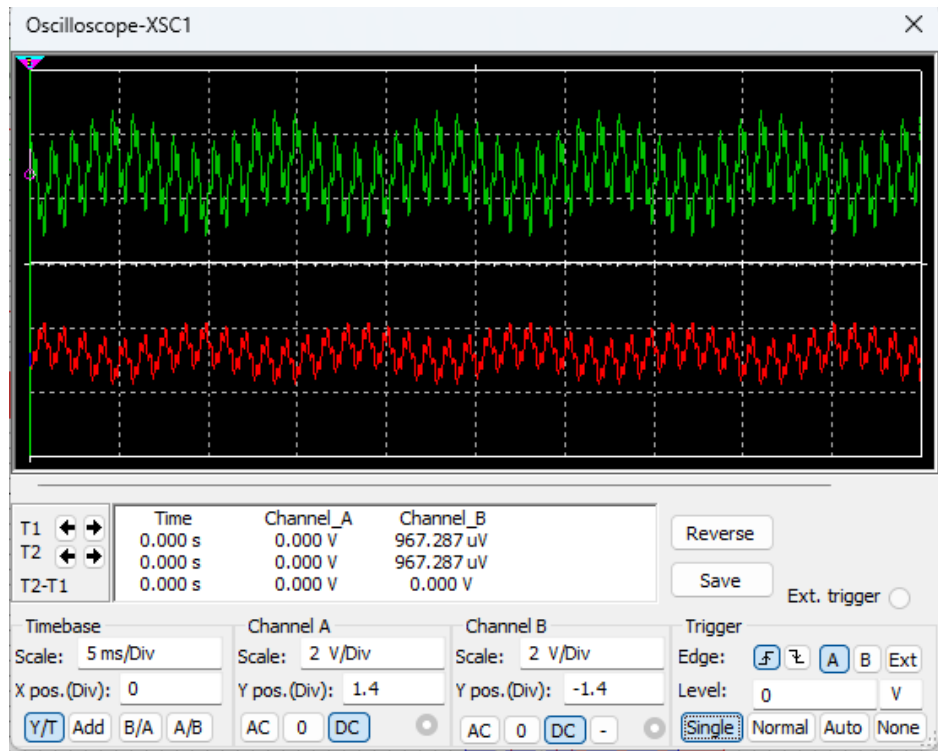


Figure 9, Block 3: Volume control

BLOCK 4: Volume Display Module

Design Objective:

The purpose of Block 4 is to create a volume display for our audio output. To implement this, we are using a resistor ladder and 4 LED's that light up when the audio amplitude exceeds 1.5V, 1V, 0.5V and 0.25V respectively.

Schematic:

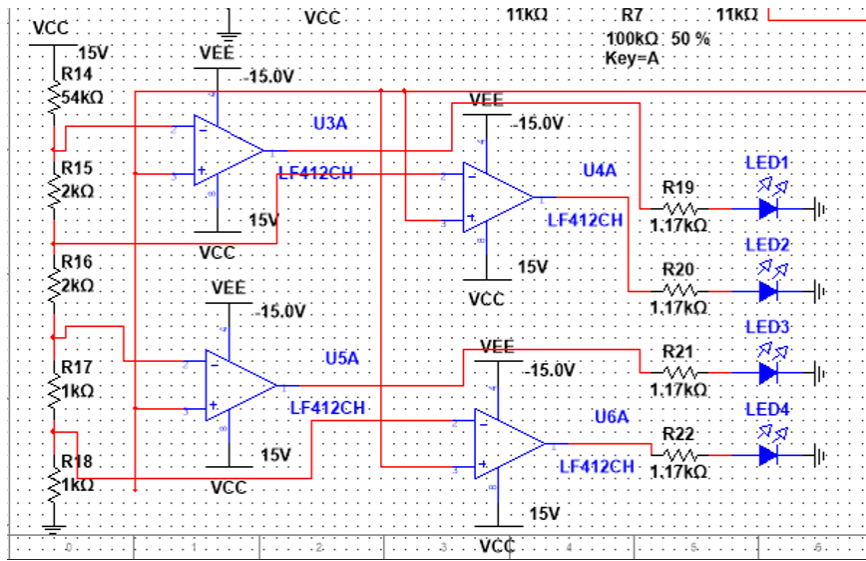


Figure 10, Schematic of Block 4.

Theory of Operation:

Block 4 works by using a resistor ladder to create reference voltages that are input into the comparator along with the audio signal. For each comparator, whenever the amplitude of the audio signal exceeds the reference voltage input into the comparator, the comparator output saturates high to drive the LED connected to it. On the other hand, if the audio signal is not able to exceed the comparator's reference voltage the comparator is saturate low and will turn the LED off.

Derivation / Analysis:

The equations for the reference voltages are:

$$V_1 = \frac{R_{15} + R_{16} + R_{17} + R_{18}}{R_{14} + R_{15} + R_{16} + R_{17} + R_{18}} \cdot 15$$

$$V_2 = \frac{R_{16} + R_{17} + R_{18}}{R_{14} + R_{15} + R_{16} + R_{17} + R_{18}} \cdot 15$$

$$V_3 = \frac{R_{17} + R_{18}}{R_{14} + R_{15} + R_{16} + R_{17} + R_{18}} \cdot 15$$

$$V_4 = \frac{R_{18}}{R_{14} + R_{15} + R_{16} + R_{17} + R_{18}} \cdot 15$$

Since we need V_1, V_2, V_3, V_4 to be 1.5V, 1V, 0.5V and 0.25V respectively, the resistor values need to be: $R_{14} = 54k\Omega$, $R_{15} = 2k\Omega$, $R_{16} = 2k\Omega$, $R_{17} = 1k\Omega$, $R_{18} = 1k\Omega$.

Simulation Results:

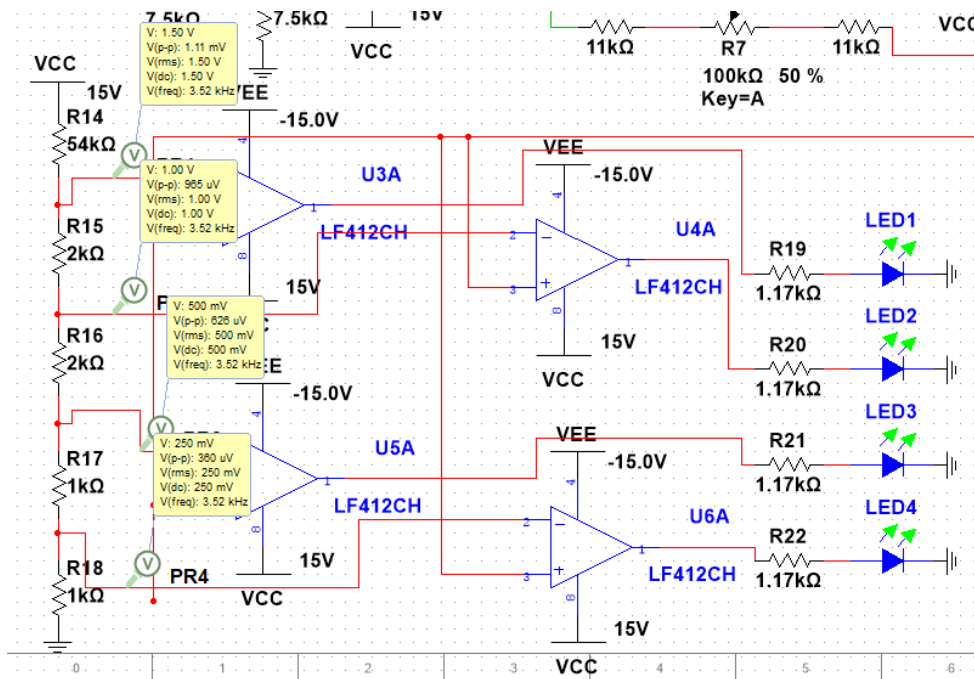


Figure 11, Block 4: Volume Display reference voltages

Note: The top channel is block input and the bottom channel is the LED signal with comparator of reference voltage 1V.

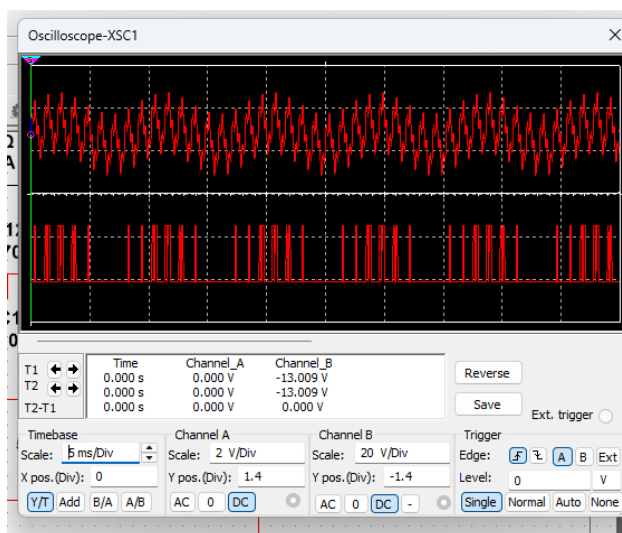


Figure 12, Block 4: Volume Display

BLOCK 5: Attenuation Module

Design Objective:

The purpose of block 5 is to attenuate the audio signal from block 3 back down to headphone safe ranges (0.5V - 1V). We need to do this because Block 1 has the capability to amplify the signal to 2 times its input and the Block 2 can amplify the signal to 10 times its input. With both combined our circuit can have an overall amplification of 20! To account for this amplification, we need to attenuate the voltage back to headphone safe levels.

Schematic:

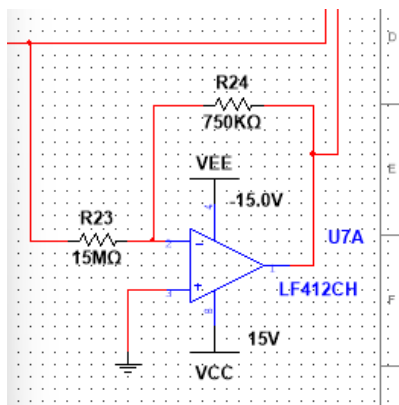


Figure 13, Schematic of block 5.

Theory of Operation:

To achieve an amplification of 1/20, we can use a simple inverting amplifier. The gain of an inverting amplifier is

$$A_v = \frac{v_o}{v_{in}} = -\frac{R_f}{R_{in}}$$

Derivation / Analysis:

We are able to choose any pair of resistors that have a ratio of 1:20, to achieve our goal, however we need to choose resistors that are considerably larger than the rest of the blocks to minimize the loading effect from those blocks. In accordance, we have chosen the resistors with values of $15\text{M}\Omega$ and $750\text{K}\Omega$.

Simulation Results:

Note: The green channel is the input to block 5 and the red channel is the output of block 5.

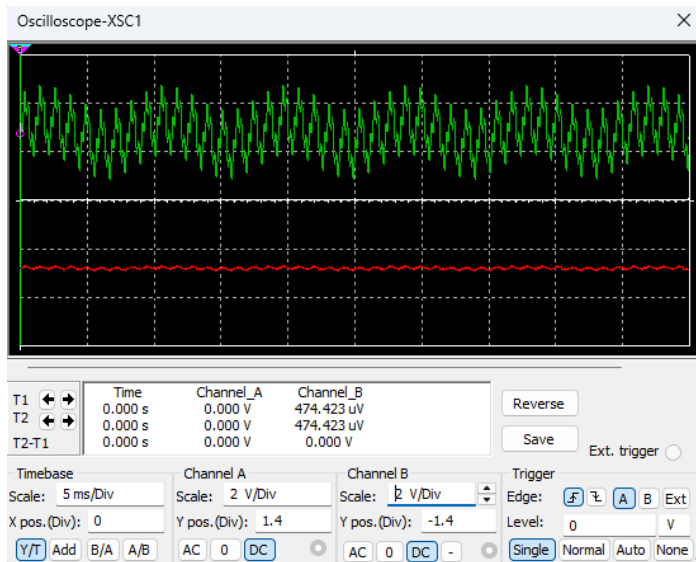


Figure 14, Block 5: Circuit Output

COMPLETE ASSEMBLY AND BREADBOARD IMAGES:

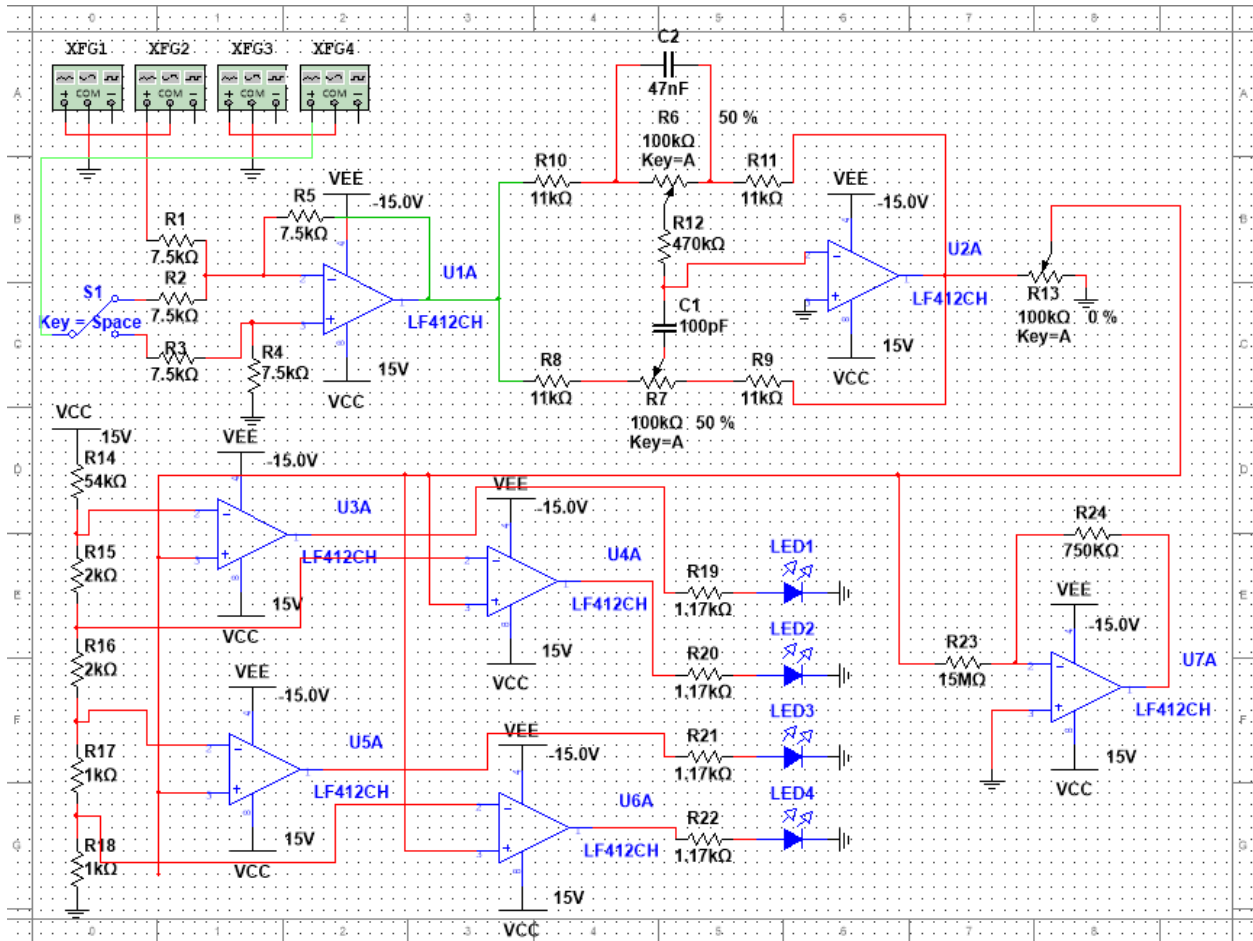


Figure 15, Complete Schematic.

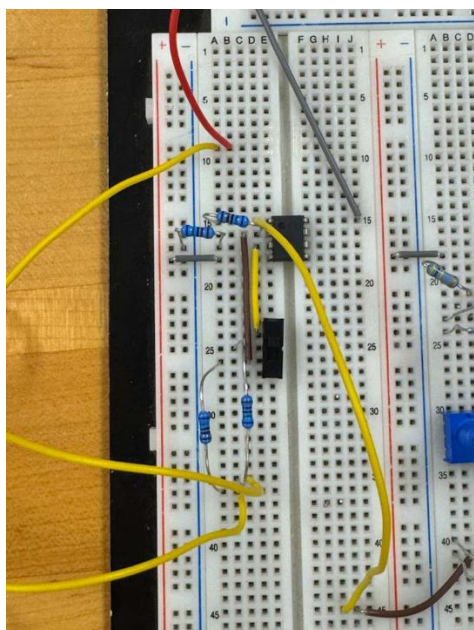


Figure 16, Block 1 on breadboard

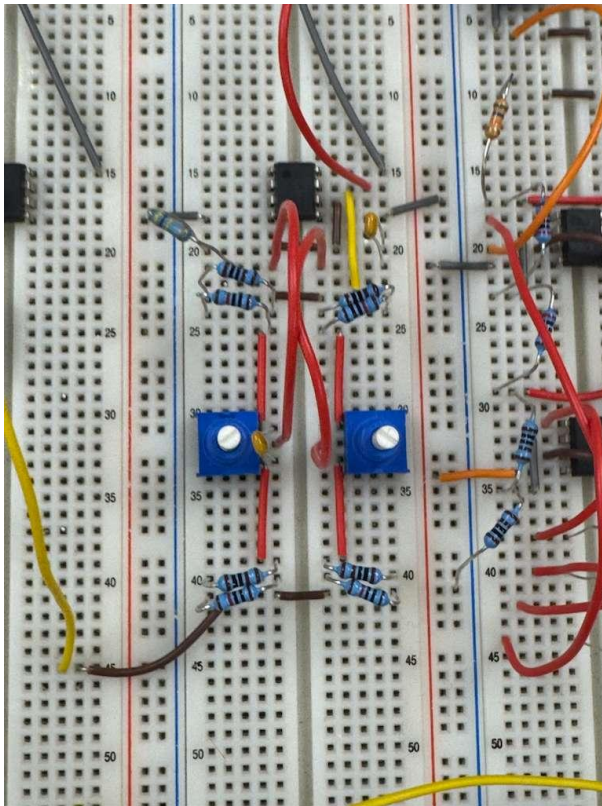


Figure 17, Block 2 on breadboard

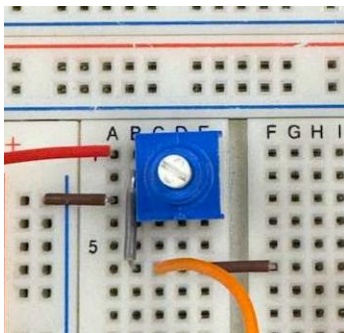


Figure 18, Block 3 on breadboard

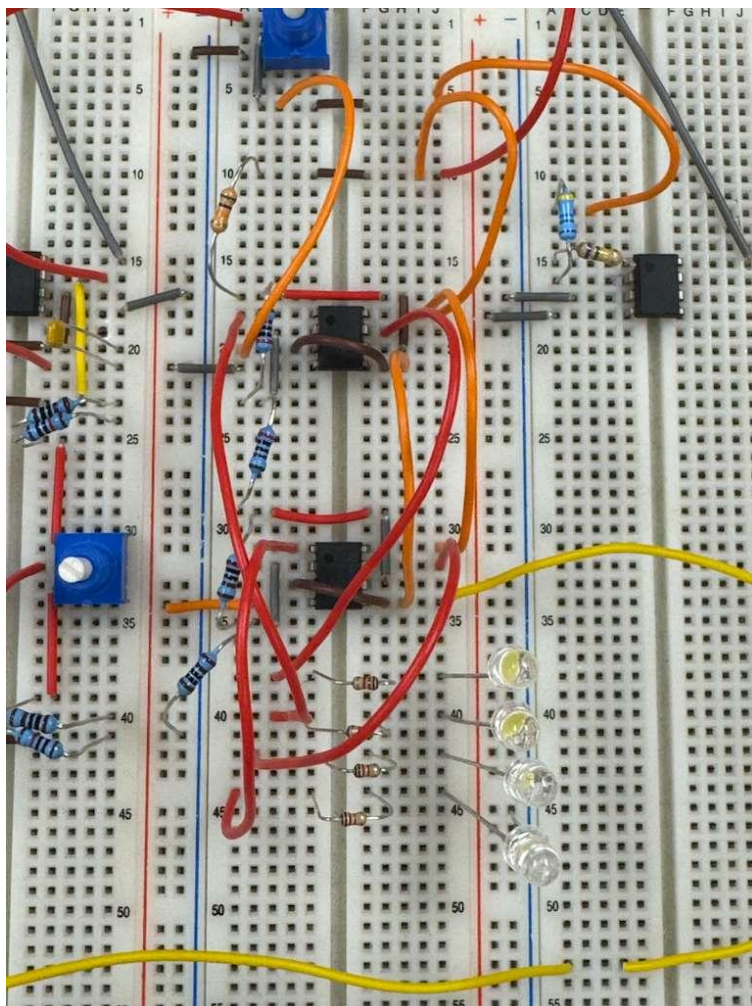


Figure 19, Block 4 on breadboard

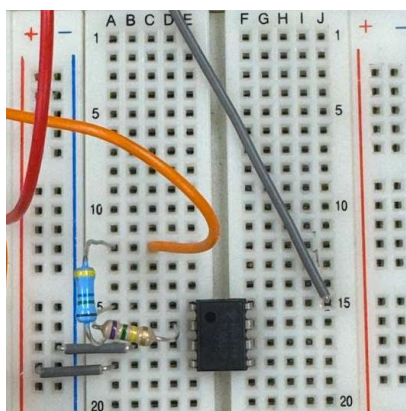


Figure 20, Block 5 on breadboard

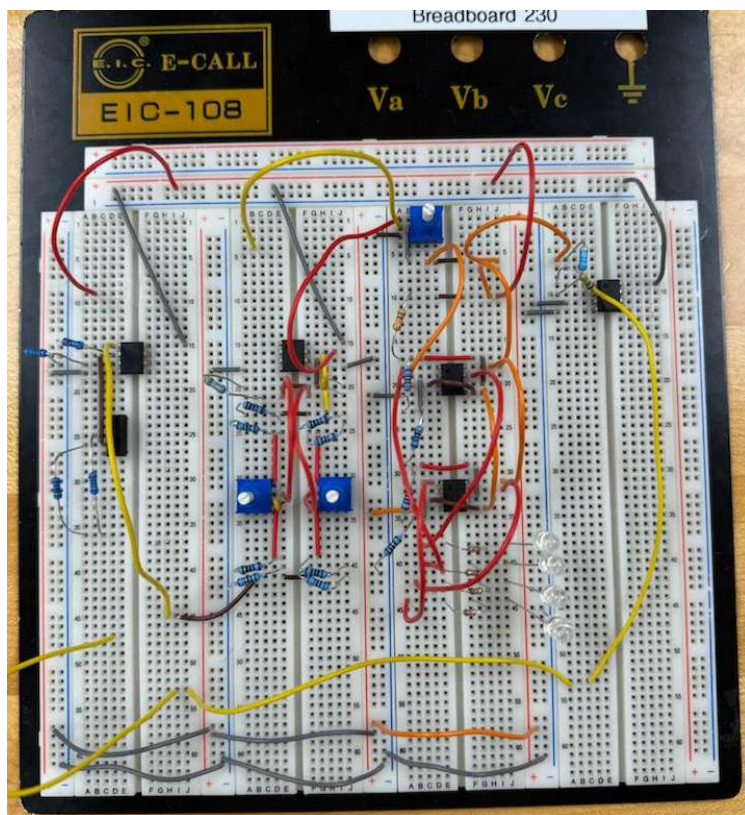


Figure 21, Overall Breadboard

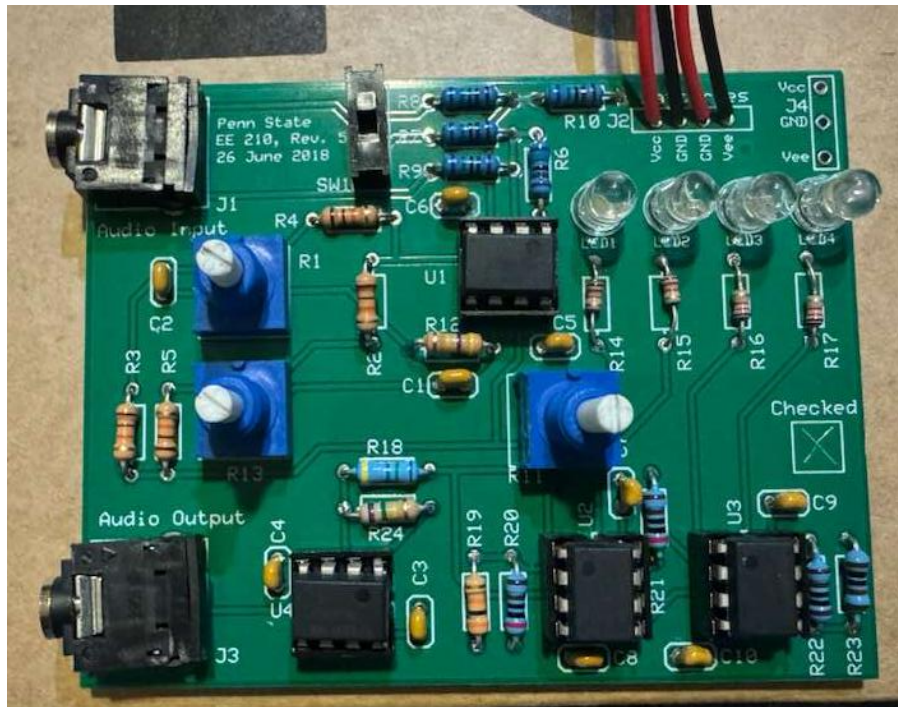


Figure 22, PCB Circuit.

Conclusion:

In this project, we were able to fully design and prototype a complete audio mixer with tone control. We individually analyzed each and block and verified the behavior with NI Multisim, after which we went on to fully confirm that our analysis was right with a real demo with a breadboard and PCB. Overall, there were some difficulties with finding the right resistor values on block 4, soldering the audio jack, loose connections on the breadboard but overall, this project taught us an incredible number of things about electrical engineering and circuit design.